

Proof. We get this by applying Proposition 6.8 both to $\text{Dep}(V)$ and to $\text{Val}(V)$, which are dual to each other by Proposition 6.4. $\square$

Therefore two (labeled) configurations of vectors or points have *the same oriented matroid* if they have the same set of circuits, or (equivalently) the same set of cocircuits, the same set of vectors, or the same set of covectors.

(c) Duality

A concept of *duality* is built into the whole structure of oriented matroids. In fact, since the vectors and the covectors of an oriented matroid $\mathcal{M}(V)$ arise in the same way as the sign vectors of a subspace, they also satisfy the same axioms (ignoring a switch from row vectors to column vectors).

Definition 6.10. The *dual* of an oriented matroid $\mathcal{M}$ is the oriented matroid $\mathcal{M}^*$ with the following properties:

- The vectors of $\mathcal{M}^*$ are the covectors of $\mathcal{M}$, and thus the circuits of $\mathcal{M}^*$ are the cocircuits of $\mathcal{M}$.
- The covectors of $\mathcal{M}^*$ are the vectors of $\mathcal{M}$, and thus the cocircuits of $\mathcal{M}^*$ are the circuits of $\mathcal{M}$.

From the way we have defined (realizable) oriented matroids, it is clear that for every oriented matroid $\mathcal{M}$, there is a unique dual oriented matroid $\mathcal{M}^*$, whose dual is

$$(\mathcal{M}^*)^* = \mathcal{M}.$$

In fact, $\mathcal{M} = \mathcal{M}(U)$ is realizable with $U \subseteq \mathbb{R}^n$, then $\mathcal{M}^* = \mathcal{M}(U^\perp)$, for the orthogonal space

$$U^\perp := \{\mathbf{c} \in (\mathbb{R}^n)^* : \mathbf{c}\mathbf{x} = 0 \text{ for all } \mathbf{x} \in U\}.$$

So existence and uniqueness of the dual follow from Proposition 6.4 (in the realizable case):

$$\mathcal{M}^* = \mathcal{M}(\text{Dep}(V)) \quad \text{for} \quad \mathcal{M} := \mathcal{M}(\text{Val}(V)).$$

Furthermore, if $\mathcal{M}$ has rank r , then $\mathcal{M}^*$ has rank $n - r$, and conversely.

(d) Deletion and Contraction

There are two very natural and fundamental operations on point configurations: *deletion* and *contraction*. They are dual to each other, and they directly translate into oriented matroid language and terminology.